API and Data Sources

Our JSON API with real examples, and the outside data a real city needs

CoolRoute: Pedestrian Routing with Shade and Heat-Exposure Budgets

1. Two kinds of API

Our API is what the app calls. **Outside data sources** feed the real-city version. The prototype needs no outside API, which is why it cannot break during a demo.

2. Our API (built and tested)

Method and path	What it does	Returns
GET /api/v1/health	Checks the server is up	status, number of nodes and edges
GET /api/v1/sun?hour=13	Sun strength for an hour	value from 0 to 1
GET /api/v1/graph	The street graph	nodes and edges with attributes
POST /api/v1/route	Shortest route and CoolRoute	both routes, explanation, warnings

Request body for POST /api/v1/route

Field	Type	Default	Meaning
origin, destination	{node} or {x, y}	required	Junction id, or metres on the local map (snapped to the nearest junction)
hour	number 0 to 24	13	Time of day
heat_budget_pct	number 10 to 100	60	Allowed heat as a percent of the shortest route's heat
step_free	true or false	false	Hard rule: use only step-free segments
shade_available	true or false	true	Set false to see the no-shade scenario
closed_edge_ids	list of ids	[]	Closed footpaths

Real request

```
{
 "origin": {
  "node": 0
 },
 "destination": {
  "node": 63
 },
 "hour": 13,
 "heat_budget_pct": 60,
 "step_free": false,
 "shade_available": true,
 "closed_edge_ids": []
}
```

Real response (route node lists left out for space)

```
{
 "sun_strength": 0.774,
 "heat_budget": 457.2,
 "budget_met": true,
 "baseline": {
  "distance_m": 1106.6,
  "walk_time_min": 15.0,
  "heat_exposure": 762.1,
  "effort": 1181.8,
  "shaded_share": 0.167,
  "data_confidence": 0.71,
  "missing_data_segments": 0,
  "segments": 14
 },
 "coolroute": {
  "distance_m": 1129.3,
  "walk_time_min": 14.6,
  "heat_exposure": 454.9,
  "effort": 1212.7,
  "shaded_share": 0.489,
  "data_confidence": 0.61,
  "missing_data_segments": 0,
  "segments": 14
 },
 "explanation": "CoolRoute is 23 m longer (+2.1%) and takes 0.4 min less, but cuts heat exposure by 40% by keeping to
 "warnings": []
}
```


Error responses (all tested)

Case	HTTP	Code	Message
Bad JSON, missing field, bad node, hour or budget	400	BAD_JSON / BAD_INPUT	Says exactly what to fix
No route with these closures or step-free rule	404	NO_ROUTE	No route exists with the closed footpaths and step-free rule given.
Budget cannot be met	200	warning BUDGET_NOT_MET	Returns the coolest route plus a warning
Large detour	200	warning LARGE_DETOUR	Cooler route is more than 20% longer
Missing shade data on the route	200	warning MISSING_DATA	Counts the weak segments

A budget that cannot be met is a **200** and not an error, because the coolest route is still useful. A missing route is a **404** because there is nothing to show.

3. Outside data sources for a real city

Need	Source	Key needed?	Notes
Walkable streets, crossings, kerbs, building outlines	OpenStreetMap, via OSMnx or the Overpass API	No	Free and open. Respect the usage policy; for a whole city, download an extract instead of calling the live API
Temperature and solar radiation	Open-Meteo forecast API	No key for non-commercial use	Gives hourly temperature and radiation, so the sun factor can use real weather. Check terms before commercial use
Tree cover (NDVI)	Sentinel-2 imagery, or an open land-cover product	Account may be needed	Free imagery. Greenness is a good stand-in for tree shade

Need	Source	Key needed?	Notes
Building heights	OSM tags (height, building:levels), or a city dataset	No	Many buildings have no height. Missing values lower confidence
Sun position	pvlib or SunCalc library	No	Replaces our simple formula with exact azimuth and elevation
Place search	Nominatim (OSM)	No	Rate limited. For production use a paid or self-hosted geocoder
Map tiles	MapLibre GL JS with an OSM-based tile provider	Depends on provider	Read the provider's free-tier terms first
Slope	A free elevation dataset such as SRTM	Account may be needed	Turn elevation into slope per street piece

> Limits and terms of free services change. Before launch, read each provider's current terms. For the hackathon, none of these is needed, because the prototype uses a built-in synthetic neighbourhood.

4. How outside data becomes a graph (production)

```
1. Download walk network for Jaipur with OSMnx          -> nodes, edges, length
2. Mark crossings and kerbs                             -> step_free, wait (default by road class)
3. Sample elevation at both ends of each edge           -> slope = rise / length x 100
4. For each hour: compute sun elevation and azimuth     -> pvlib
5. Building shadow per edge: L = H / tan(elevation)     -> shade_building (file 06)
6. Tree shade from NDVI around the edge                 -> shade_tree
7. ML model combines features                           -> shade, confidence
8. Save to PostGIS (edge_shade table)                   -> used by the router
```